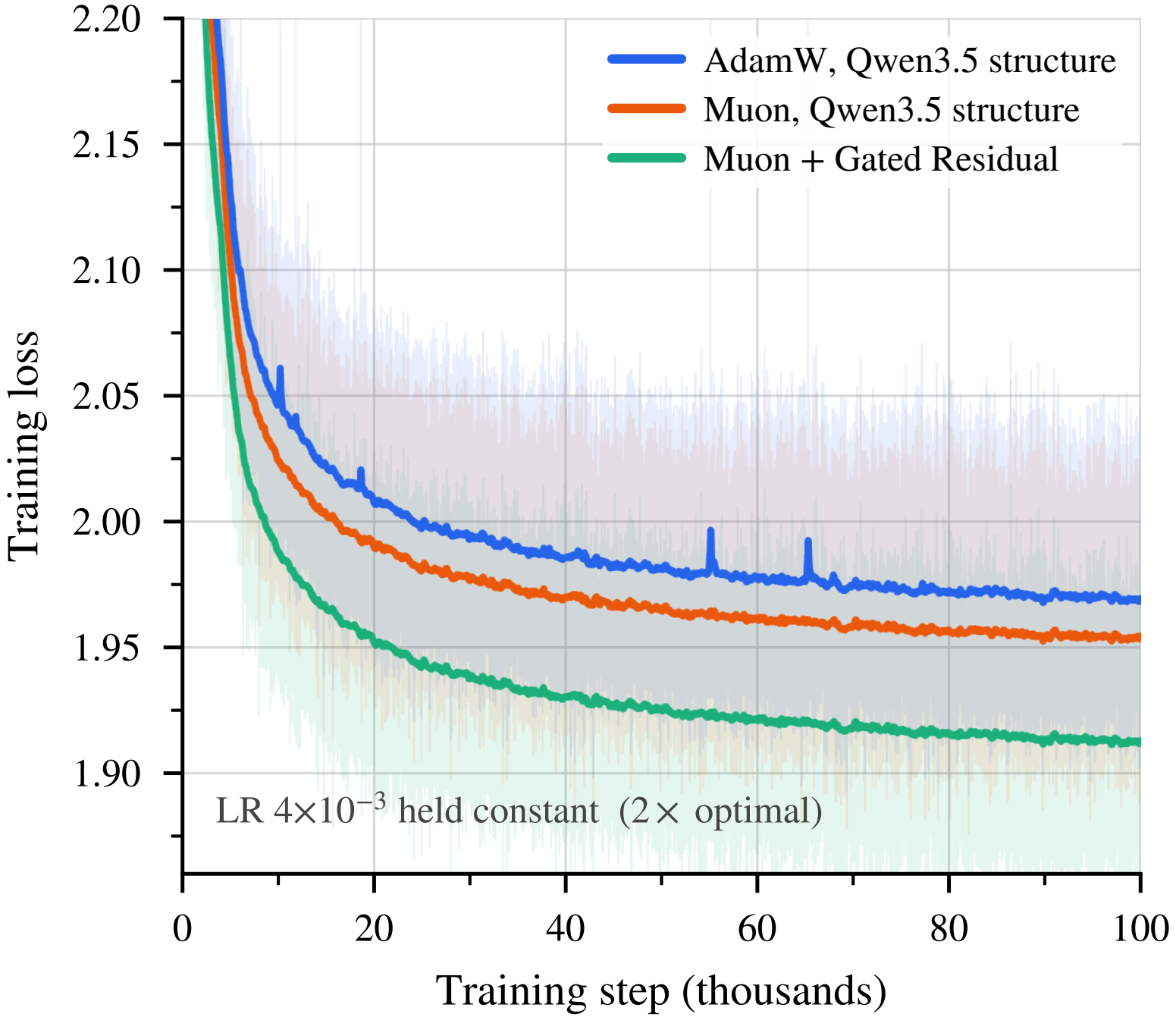

AdamW, Qwen3.5 structure
Muon, Qwen3.5 structure
Muon + Gated Residual
LR 4×10⁻³ held constant (2× optimal)
Training loss
2.20
2.15
2.10
2.05
2.00
1.95
1.90
0
20
40
60
80
100
Training step (thousands)